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#%% md
### Instructor Performance and Course Quality Evaluation on EduPro
#%% md
#### Importing all required libraries
#%%
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
#%% md
#### Reading all available datasets
#%%
df_courses = pd.read_csv("courses.csv")
df_teachers = pd.read_csv("teachers.csv")
df_transactions = pd.read_csv("transactions.csv")
df_users = pd.read_csv("users.csv")
#%% md
Let us have a look at the key analytical questions in our dataset
#%% md
#### Key Analytical Questions

1) What is the overall distribution of instructor ratings?
2) Do instructors with more experience receive higher ratings?
3) Is there a relationship between TeacherRating and CourseRating?
4) Which expertise areas consistently deliver high-quality courses?
5) Are highly rated instructors associated with higher enrollments?
#%% md
Now, let us have a look at the first question - What is the overall distribution of instructor ratings?

For this, we need to take the teachers dataset into consideration, because that is where we find the ratings of each teacher.
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df_teachers.head()
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Now, we need to check for null values and duplicates for the dataset. Null values need to be dealt with depending on the feature. Duplicates need to be removed.

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#%%
df_teachers.isna().sum()
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Now, we see that there are no null values in our dataset. We need to check for rows that contain duplicate data. Some rows can contain different teacher ID but the teacher's name, age and all remain the same. So, we should not consider teacher ID for checking duplicates.

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#%%
columns_except_teacher_id = df_teachers.columns.difference(["TeacherID"])
df_teachers[columns_except_teacher_id].duplicated().sum()
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#%% md

Now, we see that there are no duplicated rows in the dataset. This means that our dataset is clean. We can go ahead with our analysis. Now, we need to check how ratings of teachers are distributed across the dataset.

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#%%
sns.kdeplot(df_teachers["TeacherRating"])
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So, we see that ratings of all the instructors on EduPro platform are distributed almost uniformly (with slight variation between ratings 1 and 2). This means that there is a high concentration of teachers around average rating.

Let us also check how the instructors' age and experience are distributed.

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#%%
sns.kdeplot(df_teachers["Age"])
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#%% md

From the above graph, we can say that majority of

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53 teachers age between 26 and 50.
54
55 Now, let us check how well experience is distributed.
56 #%%
57 sns.kdeplot(df_teachers["YearsOfExperience"])
58 #%% md
59 The above graph says that there are teachers with
   experience of less than 0 years - it starts with -5
   years of experience, which does not make sense at all
   . Let us check our dataset again if it contains
   teachers with negative years of experience.
60 #%%
61 (df_teachers["YearsOfExperience"] < 0).sum()
62 #%% md
63 Now, we see that there are no teachers with 0 years
   of experience. Let us check why this happened.
64 #%%
65 print("Number of Teachers with no experience : ",(
   df_teachers["YearsOfExperience"]==0).sum())
66 print("Number of Teachers with one year experience
    : ",(df_teachers["YearsOfExperience"]==1).sum())
67 #%% md
68 This is because the minimum experience of teachers is
    1, and there are 7 teachers with one year of
   experience. Then until 5 years of experience, the
   number of teachers increase, and then the number of
   teachers decrease as their experience increases. This
    means that teachers with around 5 years of
   experience dominate our dataset.
69 #%%
70
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